Práctica 5 – Data Warehousing y OLAP

1. **Objetivo**

Entender los principios básicos de ***Data Warehousing*** y las operaciones ***OLAP*** por medio de extensiones **SQL**.

Pre-requisitos

- **StarUML: https://staruml.io/**
- **SGBD: PostgreSQL o Oracle o SQL Server**
  - **https://www.postgresql.org/download/**
- **Grupos de 3 a 6 integrantes**

2. **Actividades**:

La gerencia de un **hospital** quiere implantar en un ***data warehouse*** un **módulo** para almacenar información histórica sobre las **consultas (Appointment)** y visitas de sus **pacientes (Patient)** y los **médicos (Physician)** que los han atendido. El objetivo es llevar un registro del **coste** y **duración** de cada **hospitalización (stay)** correspondiente a uno o más **tratamientos (Undergoes) y procedimientos (Procedures)** que se aplican a algún **paciente**, para ello se pide:

1) Importar la **fuente de datos** proporcionada (**hospital.sql**) en el **SGBD** seleccionado.
   a. Entender Diagrama Entidad-relación y descripción de las entidades en: https://www.w3resource.com/sql-exercises/hospital-database-exercise/index.php
      https://en.wikibooks.org/wiki/SQL_Exercises/The_Hospital
   b. Importar base de datos **hospital.sql**
   c. Validar base de datos creada.
2) **Diseñar** un (ó 2) **cubo**(s) **OLAP de 4 dimensiones**:
   a. **Hechos: Medidas sobre costo** y **duración** de los **tratamientos** recibidos.
   b. **Dimensiones: médico**, **paciente** y **tiempo** (tal vez sea necesario adicionar la dimensión de **procedimiento**).
   c. De los **médicos** hay que almacenar un identificador numérico, su nombre y apellidos, y su **especialidad** (Trained_in Procedures).
   d. Los **pacientes** también tienen un identificador numérico, nombre y apellidos y, además, dirección, teléfono y número de seguro.
   e. Los **tratamientos** (**Undergoes**) se originan a partir de una **hospitalización** (**Stay**).
   f. El **cubo OLAP**, deben almacenar – además de una **referencia** (*foreign key*) a cada una de las tablas de **dimensiones** – el **costo** del **tratamiento** y su **duración** en **días**.
   g. Sobre el **tiempo** la unidad mínima de granularidad debe ser una **fecha** que incluya **hora y minuto (dd/mm/aaaa hh:mm)**, además los **periodos de tiempo** deben poder agruparse por **días**, **semanas**, **meses** y **años**.
3) Implementar el (o los) **cubo**(s) **OLAP** en la herramienta de su preferencia: **PostgreSQL o Oracle o SQL Server**
   a. Modelar el (o los) **cubo**(s) **OLAP** por medio de un **esquema estrella** o **copo de nieve** o **constelación**.
      i. **Modelo relacional**: Usar **StarUML** o similar
   b. Generar instrucciones SQL (DDL) de creación de esquema

c. Ejecutar los scripts de creación de esquema
d. En caso de falla, corregir el modelo, y volver al paso a)
e. Crear los scripts SQL (DML) necesarios para poblar las tablas de **hechos** y **dimensiones** del esquema.

```
INSERT INTO table_name (column1, column2, column3, ...)
VALUES (value1, value2, value3, ...);
```

```
INSERT INTO target_table (column1, column2, column3, ...)
SELECT column1, column2, column3, ...
FROM source_table
WHERE condition;
```

f. Ejecutar los scripts para poblar las tablas de **hechos** y **dimensiones** del esquema
g. Validar el esquema actualizado

4) Entender las **operaciones *OLAP***
   a. Usualmente, las herramientas de ***DW*** o ***BI*** utilizan el ***lenguaje MDX*** para implementar **CUBOS** y **operaciones OLAP** https://docs.microsoft.com/en-us/sql/mdx/mdx-language-reference-mdx?view=sql-server-ver15
   b. Leer Material disponible en el Aula Virtual.
   c. En esta práctica: Usar **Extensiones OLAP** para **SQL**
      i. **RANKING**:
         - https://www.tigerdata.com/learn/understanding-rank-and-dense_rank-in-postgres
      i. **GROUPING SETS, ROLLUP y CUBE**
         - **https://www.postgresql.org/docs/current/queries-table-expressions.html#QUERIES-GROUPING-SETS**
         - https://medium.com/@ebrubddl/rollup-and-cube-in-sql-4086746ec62b

5) Entender **operaciones *OLAP*** usando **SQL** y las funciones de **RANKING, GROUPING SETS, ROLLUP y CUBE.**
   a. Describir el papel de las funciones de **RANKING** (por ejemplo, **RANK()**, **DENSE_RANK()**) en **SQL**.
      i. Ordenar los **tratamientos** recibidos por los pacientes por **costo** y **duración**.
      ii. Usar **RANK()** y **DENSE_RANK()** en la consulta **SQL**, por separado**.**
      iii. Explicar los resultados de las consultas y **diferencias** entre **RANK()** y **DENSE_RANK().**
   b. Describir el papel de las funciones de **GROUPING SETS, ROLLUP y CUBE** en **SQL**. Dada una ***tabla de hechos*** con datos de **tratamientos** (por ejemplo, **hospitalizacion**) y las tablas de **dimensiones relevantes**, escriba 1 sentencia SQL para:
      i. Resumir el **costo total** de los **tratamientos** recibidos por un paciente por año y médico responsable.
      ii. **Usar GROUPING SETS, ROLLUP y CUBE** en la consulta **SQL**, por separado**.**
      iii. Explicar los resultados de las consultas y **diferencias** entre **GROUPING SETS, ROLLUP y CUBE.**
   c. Explicar las operaciones ***dice*** y ***slice*** en ***OLAP***. Dada una ***tabla de hechos*** con datos de **tratamientos** (por ejemplo, **hospitalizacion**) y las tablas de **dimensiones relevantes**, escriba 1 sentencia SQL para:

        i. segmentar (***slice***) el cubo para seleccionar solo los tratamientos del mes de mayo de 2008, y
        ii. trocearlo (***dice***) por médico.

6) Una vez diseñado e implementado el (o los) **cubo**(s) **OLAP** en la herramienta de su preferencia, diseñar consultas **SQL (JOIN; GROUP BY, RANKING, GROUPING SETS, ROLLUP, CUBE, WHERE y SUB-QUERIES)** para:
    a. Obtener el listado de pacientes con **tratamientos** de un día de duración de la primera semana del mes de mayo de 2008, ordenar el listado por nombre de paciente.
    b. Obtener el listado de los médicos que, en promedio, realizan **tratamientos** de menor duración, ordenar el listado de menor a mayor duración de los tratamientos.
    c. Obtener el coste total de todos los **tratamientos** de los médicos de la **especialidad** de "*Reverse Rhinopodoplasty*" durante el año **2008**.
    d. Listar el ranking (listado de mayor a menor) de las **especialidades** (*procedures, trained_in*) que más costes generaron durante el año 2008.
    e. Listar los pacientes que se **internaron** (*Stay*) durante el mes de mayo de 2008 y fueron atendidos por algún especialista, ordenar el listado por el total del coste de los tratamientos recibidos por dichos pacientes durante ese mes, del más caro al más barato.
7) ¿Cuáles de las consultas anteriores son implementaciones de operaciones *OLAP* en SQL?
    a. Consulta – nombre operación *OLAP*
8) Investigar e implementar la mejor forma de visualizar los resultados de las consultas.
9) Investigar y listar Herramientas de DW, OLAP o BI populares:
    a. Nombre – URL – Soporte a operaciones OLAP

3. Entregable:

- Informe sobre el diseño del cubo, esquema de DW, consultas – resultados, y reportes generados

SQL Hospital Database - Exercises, Practice, Solution

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https://www.w3resource.com/sql-exercises/hospital-database-exercise/index.php

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An introduction about hospital Database

Sample Database: hospital

With the help of a Hospital Database, this exercises will help you undestand simple SQL select queries to advanced multi table JOIN queries.

Sample Database description:

Hospitals are the most important part of our lives, trying to provide the best medical facilities to people suffering from various type of illness, which may be due to change in climate conditions, increased work-load, emotional trauma stress etc. It is very much difficult for the hospital to maintain its day-to-day acitivities and records manually. That is why a database is required to keep records of all type of acitivities of a hospital.

List of tables in the hospital database:

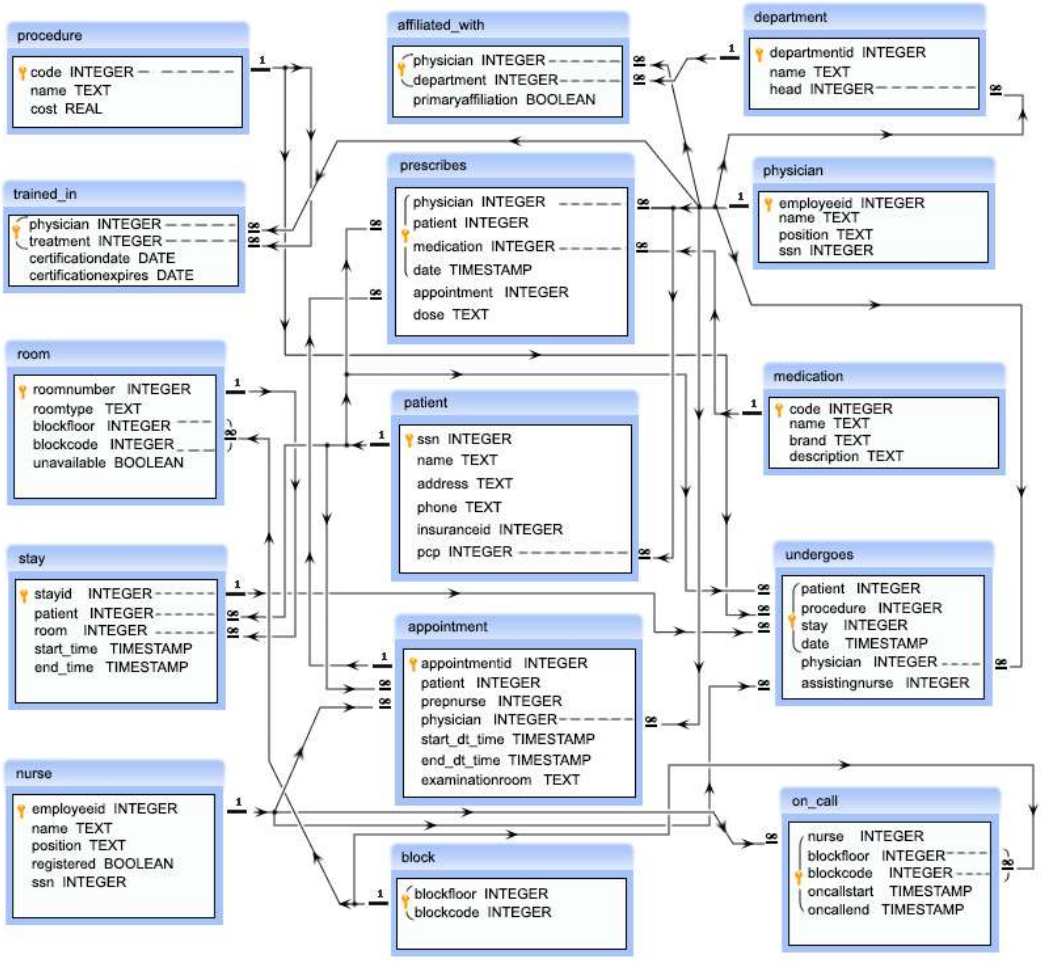
- physician
- department
- affiliated_with
- procedure
- trained_in
- patient
- nurse
- appointment
- medication
- prescribes
- block

- room
- on_call
- stay
- undergoes

Exercises

- [Basic, Subqueries, and Joins [39 exercises with solution]]

E R Diagram of Hospital Database:



physician:

- employeeid – this is a unique ID of a physician
- name – this is the name of a physician
- position – this is the designation of a physician

- ssn – this is a security number of a physician

department:

- departmentid – this is a unique ID for a department
- name – this is the name of a department
- head – this is the ID of the physician who is the head of a department, referencing to the column employeeid of the table physician

affiliated_with:

- physician – this is the ID of the physicians which is referencing to the column employeeid of the physician table
- department – this is the ID the department which is referencing to the column departmentid of the department table
- primaryaffiliation – this is a logical column which indicate that whether the physicians are yet to be affiliated or not
- *Note: The combination of physician, department will come once in that table.*

procedure:

- code – this is the unique ID of a medical procedure
- name – the name of the medical procedure
- cost – the cost for the procedure

trained_in:

- physician – this is ID of the physicians which is referencing to the column employeeid of the physician table
- treatment – this is the ID of the medical procedure which is referencing to the column code of the procedure table
- certificationdate – this is the starting date of certification
- certificationexpires – this is the expiry date of certification
- *Note: The combination of physician and treatement will come once in that table.*

patient:

- ssn – this is a unique ID for each patient
- name – this is the name of the patient
- address – this is the address of the patient
- phone – this is the phone number of the patient

- insuranceid – this is the insurance id of the patient
- pcp – this is the ID of the physician who primarily checked up the patient which is referencing to the column employeeid of the physician table

nurse:

- employeeid – this is the unique ID for a nurse
- name – name of the nurses
- position – the designation of the nurses
- registered – this is a logical column which indicate that whether the nurses are registered for nursing or not
- ssn – this is the security number of a nurse

appointment:

- appointmentid – this is the unique ID for an appointment
- patient – this is the ID of each patient which is referencing to the ssn column of patient table
- prepnurse – the ID of the nurse who may attend the patient with the physician, which is referencing to the column employeeid of the nurse table
- physician – this is the ID the physicians which is referencing to the employeeid column of the physician table
- start_dt_time – this is the schedule date and approximate time to meet the physician
- end_dt_time – this is the schedule date and approximate time to end the meeting
- examinationroom – this the room where to meet a patient to the physician

medication:

- code – this is the unique ID for a medicine
- name – this is the name of the medicine
- brand – this is the brand of the medicine
- description – this is the description of the medicine

prescribes:

- physician – this is the ID of the physician referencing to the employeeid column of the physician table
- patient – this is the ID of the patient which is referencing to the ssn column of the patient table

- medication – the ID of the medicine which is referencing to the code of the medication table
- date – the date and time of the prescribed medication
- appointment – the prescription made by the physician to a patient who may taken an appointment which is referencing to column appointmentid of appointment table
- dose – the dose prescribed by the physician
- *Note: The combination of physician, patient, medication, date will come once in that table.*

block:

- blockfloor – ID of the floor
- blockcode - ID of the block
- *Note: The combination of blockfloor, blockcode will come once in that table.*

room:

- roomnumber – this is the unique ID of a room
- roomtype – this is type of room
- blockfloor - this is the floor ID where the room in
- blockcode – this is the ID of the block where the room in
- unavailable – this is the logical column which indicate that whether the room is available or not
- *Note: The of blockfloor, blockcode columns are refercing to the combination of blockfloor and blockcode columns of the table block.*

on_call:

- nurse – this is ID of the nurse which is referencing to the employeeid column of the table nurse
- blockfloor - this is the ID of the floor
- blockcode – this is the ID of block
- oncallstart - the starting date and time of on call duration
- oncallend – the ending date and time of on call duration
- *Note: The combination of nurse, blockfloor, blockcode, oncallstart, oncallend will come once in that table and the combination of blockfloor, blockcode columns are refercing to the combination of blockfloor and blockcode columns of the table block .*

stay:

- stayid - this is unique ID for the admission
- patient – this is the ID of the patient which is referencing the ssn column of patient table
- room - this is the ID of the room where the patient admitted and which is referencing to the roomnumber column of the room table
- start_time – this is the time when a patient admitted
- end_time – this is the time how long a patient is staying

undergoes:

- patient - this is ID of the patient which is referencing to the ssn column of the patient table
- procedure – this is ID of the procedure and referencing to the code column of the procedure table
- stay - this is the ID admission of a patient, which is referencing to the stayid column of the stay table
- date – this is the date when a patient undergoes for a medical procedure
- physician – this is the ID of a physician which is referencing to the column employeeid of the table physician
- assistingnurse – this is the ID of a nurse who will assists the physician, referencing to the column employeeid of the table nurse
- *Note: The combination of patient, procedure, stay, date will come once in that table.*

The database for hospital management system used for this exercises is based upon a database available in wikipedia. Sturcture and Data are may have gone through alterations sporadically.